SentiBias Audit

Auditing Multilingual Sentiment Models for Dialect, Gender and Code-Switching Bias

AI Project — Deliverable 1: Problem Definition & Data Foundations

Saad Jahangir Talmeez ur Rehman Muhammad Yaseen Asif

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Abstract

SentiBias Audit is a research project (Track B) that tests multilingual sentiment models for bias across dialects, gender-coded names, and code-switched Roman Urdu–English text, and then applies mitigation methods to reduce the measured performance gaps. This first deliverable defines the problem (Phase 1) and builds a working, data-centric pipeline (Phase 2): three public datasets are collected with logged download scripts, cleaned (missing values, duplicates, outliers), normalised, encoded and scaled, and explored through EDA charts. All code is in the `data_scripts/` folder of the repository.

1 Phase 1: Problem Definition

1.1 Problem Statement

Sentiment analysis models are deployed at scale for content moderation, brand monitoring, customer support triage and social-media analytics. Most of these models are trained and evaluated on standard English, yet their outputs are applied to users who write in other language varieties. Two groups are especially affected in Pakistan and the wider South-Asian diaspora:

- **Roman Urdu and code-switched writers.** A very large share of informal online Urdu is written in Latin script ("Roman Urdu") with no standard spelling (*nahi / nhi / nahin*) and is frequently mixed with English words within a single sentence. Models that have never seen this variety tend to default to *neutral* or produce unstable predictions, so the opinions of these users are systematically mis-measured.
- **People referred to by gender-coded names.** Prior work (Kiritchenko & Mohammad, 2018) showed that many sentiment systems assign different sentiment intensity to two sentences that differ only in a first name (*"Amanda feels angry"* vs. *"Alonzo feels angry"*). Such differences are not caused by the text's meaning and constitute an unfair, measurable bias.

The real-world pain point is therefore that a sentiment model can be *accurate on average* while being *unreliable for specific language varieties and demographic groups.* These gaps are invisible in a single accuracy number and are rarely reported by model providers. SentiBias Audit quantifies these gaps on multilingual transformer sentiment models and tests whether standard mitigation methods (data augmentation, re-weighting, counterfactual fine-tuning) reduce them.

1.2 Target Audience

- **Research gap.** Bias audits of sentiment analysis exist for English gender/race (EEC) and for a few high-resource languages, but there is no systematic audit that combines (i)

Roman Urdu, (ii) Roman Urdu–English code-switching and (iii) gender-coded names, nor one that evaluates mitigation methods on all three dimensions together. This project fills that gap for the low-resource Roman Urdu setting.

- **Practitioners.** NLP engineers and product teams in Pakistan deploying off-the-shelf multilingual sentiment models (e.g. for Urdu-speaking customers) who need to know *where* a model fails before trusting its output.
- **Fairness researchers and educators** who need a reproducible pipeline and a reusable, cleaned evaluation set for South-Asian code-switched text.

1.3 AI Justification

A rule-based or lexicon-based approach is not viable for this problem for four technical reasons:

1. **Open vocabulary with no orthographic standard.** Roman Urdu has no fixed spelling; the same word appears in dozens of surface forms (*acha, achha, acha, aCha*). A hand-written lexicon can never enumerate these variants, whereas sub-word tokenisation and learned embeddings in neural models generalise across spellings from data.
2. **Context-dependent polarity.** Sentiment is compositional (*"bilkul bhi acha nahi"* = not good at all). Negation, sarcasm and intensifiers interact in ways that require learned contextual representations (transformers) rather than fixed rules.
3. **Code-switching.** Within one sentence the language can change at any token boundary. A rule system would need separate language identification, separate lexicons and merging logic; a multilingual deep model handles the mixed sequence natively.
4. **The object of study is a learned model.** The bias we audit emerges from the statistical patterns a model absorbs from training data (e.g. name–sentiment associations). Measuring and mitigating it (re-weighting, counterfactual augmentation, fine-tuning) is intrinsically a machine-learning problem; there is no rule-based counterpart.

Hence Machine Learning—specifically pre-trained multilingual transformers fine-tuned for sentiment classification—is necessary both as the subject of the audit and as the means of mitigation.

2 Phase 2: Data Acquisition & Preprocessing

2.1 Collection

Three public datasets were selected, one per bias dimension (Table 1). The script `data_scripts/download_data.py` fetches every file over HTTP with the `requests` library, unzips the EEC archive, and appends one line per file (timestamp, HTTP status, byte count, URL) to the log file `data/raw/download_log.txt` as evidence of collection. Raw data is git-ignored and is regenerated by re-running the script.

Dataset	Bias dimension	Raw rows	Source
Roman Urdu Data Set	Roman Urdu, code-switching	20,229	UCI ML Repository #458
TweetEval Sentiment	English baseline (tweets)	59,899	Cardiff NLP, GitHub
Equity Evaluation Corpus	Gender-coded names	8,640	Kiritchenko & Mohammad 2018

Table 1: Datasets collected in Phase 2.

Excerpt of the download log produced by the script:

```
[2026-10-24 12:41:33] OK    roman_urdu.csv <- https://archive.ics.uci.edu/...
                            (HTTP 200, 1628349 bytes)
[2026-10-24 12:41:37] OK    tweeteval_train_text.txt <- https://raw.githubusercontent.com/...
                            (HTTP 200, 4970029 bytes)
[2026-10-24 12:41:56] OK    eec.zip <- https://saifmohammad.com/WebDocs/EEC/...
                            (HTTP 200, 1669592 bytes)
[2026-10-24 12:41:56] UNZIP eec.zip -> ['Equity-Evaluation-Corpus/...csv', ...]
```

Raw data observations. The Roman Urdu CSV has no header, a third column that is empty except for 7 junk cells, one row with no text, 564 duplicated texts, and a typo label "`Neative`" (1 row). TweetEval ships text and labels in separate aligned files with integer labels (0/1/2) and contains 28 duplicate tweets. EEC has no duplicates but 2,880 missing `Race` values (rows that use noun phrases such as *"my son"* instead of names) and 240 missing `Emotion` values (neutral templates). Figure 1 shows the raw class distributions, including the typo label.

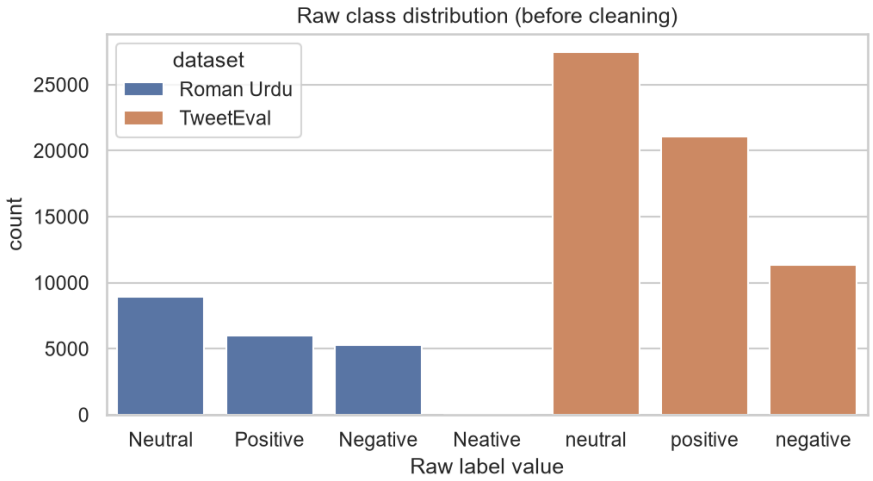


Figure 1: Raw class distribution of the two labelled corpora before cleaning. The "Neative" bar (1 row) is the label typo found in the Roman Urdu file.

2.2 Cleaning Implementation

The script `data_scripts/clean_data.py` applies the same three steps to each dataset and writes one tidy CSV per dataset plus a combined `sentiment_master.csv` with the unified schema

```
text, label, language, source, split, gender, code_switched,
n_tokens, n_chars, english_word_ratio
```

A JSON summary of every step is saved to `data/processed/cleaning_summary.json`.

1. **Missing values.** Rows with a null text or label are dropped; whitespace is stripped and collapsed and empty strings removed. The junk third column of the Roman Urdu file is dropped and the `Neative` label is corrected to `Negative`. In EEC the missing `Race`/`Emotion` cells are semantically meaningful, so they are imputed with the category `none` rather than dropped; the emotion is then mapped to a sentiment label (joy → positive; anger, sadness, fear → negative; none → neutral).

2. **Duplicates.** Exact duplicate texts are removed, keeping the first occurrence.

3. **Outliers.** Sample length in tokens is the main outlier dimension (the raw Roman Urdu file ranges from 1 to 2,017 characters). Rows above Tukey's outer fence $Q_3 + 3\,\mathrm{IQR}$ on

token count (fence = 50 tokens for Roman Urdu, 43 for TweetEval, 13 for EEC) or with zero tokens are removed.

Dataset	Raw	After missing	After duplicates	After outliers
Roman Urdu	20,229	20,116	19,582	19,187
TweetEval	59,899	59,899	59,869	59,869
EEC	8,640	8,640	8,640	8,640
Master	88,768			**87,696**

Table 2: Row counts after each cleaning step.

Code-switching flag. For the Roman Urdu corpus a lexicon heuristic marks a sample as code-switched when it contains at least two words from a curated list of unambiguous English words (ambiguous tokens such as *a, to, me, he, is* that are also valid Roman Urdu words are excluded) and is not entirely English. 959 samples (5.0%) are flagged, which forms the code-switched evaluation subset for the audit.

2.3 Feature Strategy

The script `data_scripts/feature_eda.py` implements the feature pipeline and saves the outputs to `data/processed/`:

Normalisation (text). Lowercasing; URLs replaced by `URL`; user mentions by `@user`; digits by `NUM`; punctuation removed; whitespace collapsed. Stored as `text_norm`.

Encoding. The target is label-encoded with `sklearn.LabelEncoder` (negative = 0, neutral = 1, positive = 2). The categorical variables `language` and `gender` are one-hot encoded (5 columns). The text is encoded as a TF-IDF matrix (unigrams + bigrams, 5,000 features, `min_df`=2) and saved as `features_tfidf.npz`; this is the baseline representation used before moving to transformer embeddings in later deliverables.

Scaling. The numeric features `n_tokens`, `n_chars` and `english_word_ratio` are scaled twice: z-score standardisation (`StandardScaler`, suffix `_std`) for distance/linear models and min–max scaling to $[0, 1]$ (`MinMaxScaler`, suffix `_minmax`) for neural inputs. The scaled columns, one-hot columns and encoded label form `features_tabular.csv` (13 columns, 87,696 rows).

2.4 Exploratory Data Analysis

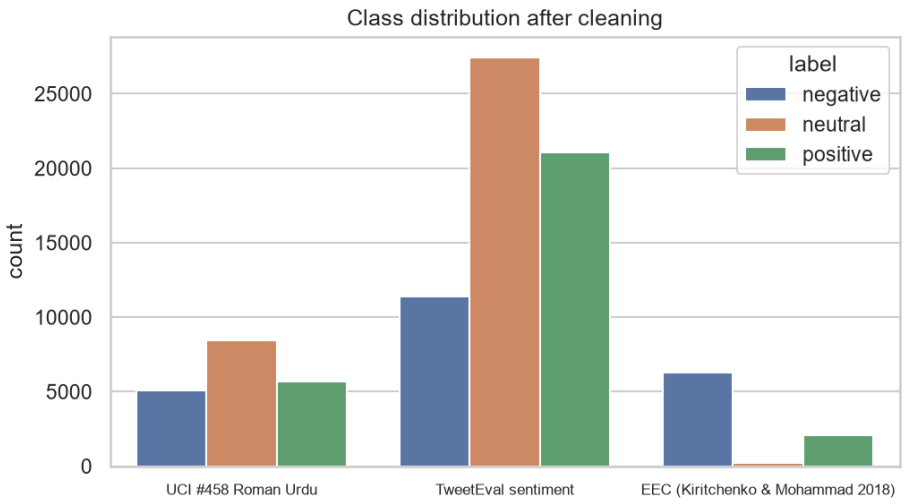


Figure 2: Class distribution per source after cleaning. Neutral dominates both the Roman Urdu and TweetEval corpora; EEC is negative-heavy by construction (three of its four emotions are negative), which will need to be accounted for when reporting per-group accuracy.

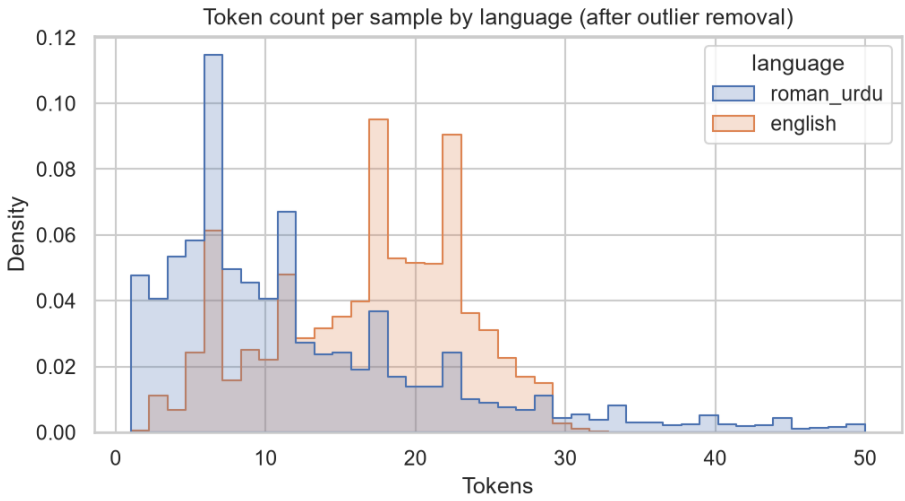


Figure 3: Token-count density by language. Roman Urdu samples are short (median 9 tokens) and skewed, whereas English tweets cluster around 18–23 tokens. This length mismatch is a potential confound for cross-language comparisons.

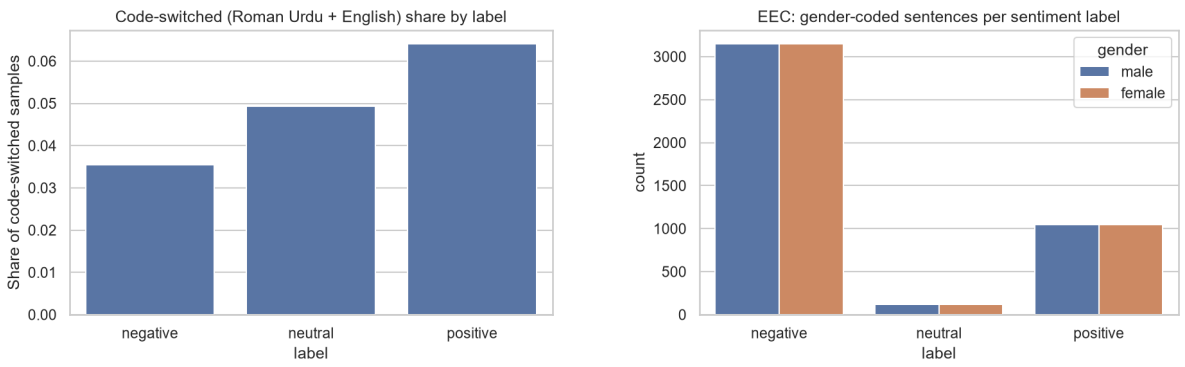


Figure 4: Left: share of code-switched Roman Urdu samples by label—positive samples switch to English almost twice as often as negative ones (6.4% vs. 3.5%), so code-switching is correlated with the label and must be controlled for. Right: EEC is exactly balanced between male and female names within every label, which is what makes paired counterfactual evaluation possible.

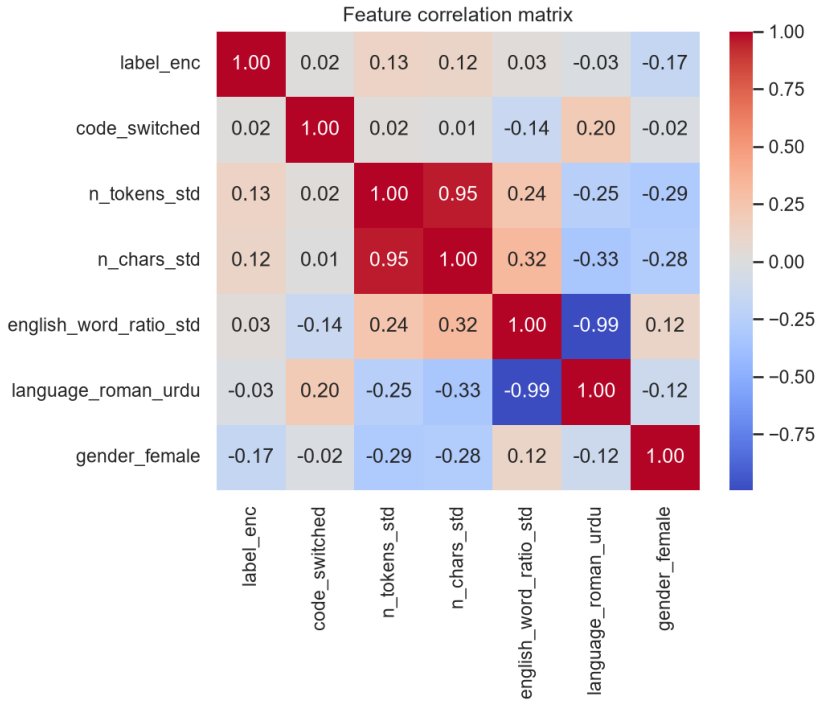


Figure 5: Pearson correlation of the tabular features. `n_tokens` and `n_chars` are nearly collinear (0.95), so only one will be used as a covariate. The −0.99 between `english_word_ratio` and `language_roman_urdu` is expected because the ratio is constant (1.0) for English rows. The label is only weakly related to length (0.13) and to the female-name indicator (−0.17, driven by EEC's label composition).

3 Repository Summary

The GitHub repository contains `README.md`, a `.gitignore` that excludes all raw and processed data, the `data_scripts/` folder with the three Phase 2 scripts (`download_data.py`, `clean_data.py`, `feature_eda.py`), `requirements.txt`, and this report with its figures under `report/`. Running the three scripts in order reproduces every number and chart in this document.

References

[1] Sharf, Z. and Rahman, S. U. (2018). *Roman Urdu Data Set*. UCI Machine Learning Repository, dataset 458. https://archive.ics.uci.edu/dataset/458/roman+urdu+data+set

[2] Barbieri, F., Camacho-Collados, J., Espinosa-Anke, L. and Neves, L. (2020). TweetEval: Unified Benchmark and Comparative Evaluation for Tweet Classification. *Findings of EMNLP 2020*. https://github.com/cardiffnlp/tweeteval

[3] Kiritchenko, S. and Mohammad, S. M. (2018). Examining Gender and Race Bias in Two Hundred Sentiment Analysis Systems. *Proceedings of *SEM 2018*. https://saifmohammad.com/WebPages/Biases-SA.html